



## CIE-II Second Semester MCA

Subject Name: Advanced Design and Analysis of Algorithms Subject Code: 25MCA230  
Time: 1 hr Max. Marks: 20

### Course Outcomes:

- CO1: Design new algorithms, prove them correct, and analyze their asymptotic and absolute runtime and memory demands.  
CO2: Find an algorithm to solve the problem (create) and prove that the algorithm solves the problem correctly (validate).  
CO3: Understand the mathematical criterion for deciding whether an algorithm is efficient and know many practically important problems that do not admit any efficient algorithms.  
CO4: Apply classical sorting, searching, optimization and graph algorithms.  
CO5: Understand basic techniques for designing algorithms, including the techniques of recursion, divide-and-conquer, and greedy.

**PART A: ANSWER ALL THE QUESTIONS** Each question carries 1mark 4Qx1=04

| Q | QUESTIONS                                    | BL | CO |
|---|----------------------------------------------|----|----|
| 1 | a Define a Trie.                             | 1  | 2  |
|   | b State the main idea of Divide and Conquer. | 1  | 3  |
|   | c Define Hamiltonian Cycle with example.     | 2  | 4  |
|   | d Define Dynamic Programming with example?   | 2  | 4  |

**PART B: ANSWER ALL THE QUESTIONS** Each question carries 4 marks 3QX4=12

| Q  | QUESTIONS                                                                                                                                                                                | BL | CO |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| 2  | a Explain Skip List structure and its working. Prove that skip list height is $O(\log n)$ .                                                                                              | 3  | 2  |
|    | OR                                                                                                                                                                                       |    |    |
| 3  | b Compute step-by-step solution for Convex Hull with the following points using divide and conquer method.<br>P1(0,0), P2(1,1), P3(2,2), P4(0,3), P5(3,0), P6(3,3)                       | 3  | 3  |
|    | a A truck has capacity 50 kg.<br><br>Item Profit Weight<br>1 60 10<br>2 100 20<br>3 120 30<br><br>Solve the problem using Fractional Knapsack. Why does greedy fail in the 0/1 knapsack? | 4  | 3  |
| OR |                                                                                                                                                                                          |    |    |



|   |                                                                                                                                                    |                                                                                 |   |   |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---|---|
| 4 | b                                                                                                                                                  | Differentiate between Prim's and Kruskal Algorithm                              |   |   |
|   |                                                                                                                                                    | Edges:<br>A-B=2, A-C=3, B-C=1, B-D=4, C-D=5<br>Find MST using Prim's Algorithm. | 3 | 3 |
|   | a                                                                                                                                                  | Explain Backtracking vs Branch and Bound with examples.                         | 3 | 4 |
| b | OR                                                                                                                                                 |                                                                                 |   |   |
|   | <p>Consider the directed weighted graph and find the shortest path distance between every pair of vertices using the Floyd-Warshall Algorithm.</p> |                                                                                 |   |   |

**PART C: ANSWER ANY 1 QUESTION**

Each question carries 4 marks

1Qx4=4

| Q | QUESTIONS                                                                                                                                                                                                                                                                                                                                     | BL | CO |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| 5 | <p>A bank maintains customer transaction records in sorted order. Frequent insertions and deletions occur. Insert the following transaction IDs and show the balancing, rotation, and recolouring in the <b>Red-Black tree</b>:</p> <p>50, 30, 70, 20, 40, 60, 80</p> <p>Also analyses the time and space complexities of this operation.</p> | 4  | 2  |
| 6 | <p>A GPS system computes shortest path.</p> <p>Edges:<br/>A-B=4, A-C=1, C-B=2, B-D=1, C-D=5</p> <p>Solve using <b>Dijkstra's Algorithm</b>. Why does Dijkstra fail with negative weights?</p>                                                                                                                                                 | 4  | 3  |





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|                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO1: Design new algorithms, prove them correct, and analyze their asymptotic and absolute runtime and memory demands.                                                              |
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| CO3: Understand the mathematical criterion for deciding whether an algorithm is efficient and know many practically important problems that do not admit any efficient algorithms. |
| CO4: Apply classical sorting, searching, optimization and graph algorithms.                                                                                                        |
| CO5: Understand basic techniques for designing algorithms, including the techniques of recursion, divide-and-conquer, and greedy.                                                  |

#### PART A: ANSWER ALL THE QUESTIONS

Each question carries 1 mark 4Qx1=04

| Q | QUESTIONS                                                                       | BL | CO |
|---|---------------------------------------------------------------------------------|----|----|
| 1 | a Define an algorithm. Why is it important?                                     | 1  | 1  |
|   | b Discuss Omega ( $\Omega$ ) notation? How is it different from Big-O notation? | 2  | 1  |
|   | c Compare binomial heap and Fibonacci heap.                                     | 2  | 2  |
|   | d State the Min-heap property with an example.                                  | 1  | 2  |

#### PART B: ANSWER ALL THE QUESTIONS

Each question carries 4 marks 3QX4=12

| Q | QUESTIONS                                                                                                                                                                                  | BL | CO |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| 2 | a Analyze the significance of asymptotic notations in comparing algorithms. Explain the difference between best-case, average-case, and worst-case complexity.                             | 3  | 1  |
|   | OR                                                                                                                                                                                         |    |    |
|   | b Explain the gap sequence in Shell Sort? How does it affect performance? Show with an example.                                                                                            | 3  | 1  |
| 3 | a Write the partition algorithm used in Quick Sort. Analyze the best case, worst case, and average case complexity of Quick Sort.                                                          | 3  | 1  |
|   | OR                                                                                                                                                                                         |    |    |
|   | b Explain the Merge Sort algorithm with an example. Discuss why Merge Sort has $O(n \log n)$ complexity in all cases.                                                                      | 3  | 1  |
| 4 | a Discuss linear time sorting algorithms and discuss when they outperform comparison-based sorting algorithms. Also, provide their comparative analysis based on complexity and stability. | 3  | 1  |
|   | OR                                                                                                                                                                                         |    |    |
|   | b Define a Fibonacci Heap with an example. List the main operations supported by a Fibonacci Heap. Write the algorithm for the Fibonacci Heap Insert operation.                            | 3  | 2  |



JSS MAHAVIDYAPEETHA  
JSS UNIVERSITY, NOIDA



PART C: ANSWER ANY 1 QUESTION

Each question carries 4 marks

1Qx4=4

| Q | QUESTIONS                                                                                                                                                                                                                                                                                                                          | BL | CO |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| 5 | <p>A modern file system stores file metadata using a B-tree index to efficiently manage and retrieve file records. Each node of the B-tree contains file identifiers (keys) sorted in ascending order. The file system uses a B-tree of order 4. Insert the following file IDs sequentially:</p> <p>10, 20, 30, 40, 50, 60, 70</p> | 4  | 2  |
| 6 | <p>Define binomial heap with its properties? State the algorithm of the extract_min operation. If a binomial heap contains 13 elements, determine the possible binomial trees present in the heap.</p>                                                                                                                             | 3  | 2  |

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